

Vaccination Policy under Model Uncertainty

Can the Needs of the Few Outweigh the Needs of the Many?

Jürgen Landes Ignacio Ojea Quintana

Munich Center for Mathematical Philosophy
LMU Munich
juergen_landes@yahoo.de

June 26, 2026

“The needs of the many outweigh the needs of the few.”

Spock, **The Wrath of Khan**

“The needs of the many outweigh the needs of the few.”

Spock, **The Wrath of Khan**

This talk asks when that intuition **fails**.

Outline

1 The Orthodoxy

2 The Claim

3 The Model

4 Results

5 Objection & Humility

6 Conclusion

The Orthodoxy

The pro-vaccination consensus

Vaccination campaigns have had spectacular successes (smallpox, polio, tetanus, COVID-19), also through **herd immunity**.

The pro-vaccination consensus

Vaccination campaigns have had spectacular successes (smallpox, polio, tetanus, COVID-19), also through **herd immunity**.

This success underwrote a strong ethical claims during COVID-19:

- Vaccine refusal is **free-riding** [1, 2];
- Mass vaccination would **surely** improve public health.

The pro-vaccination consensus

Vaccination campaigns have had spectacular successes (smallpox, polio, tetanus, COVID-19), also through **herd immunity**.

This success underwrote a strong ethical claims during COVID-19:

- Vaccine refusal is **free-riding** [1, 2];
- Mass vaccination would **surely** improve public health.

“Transmission models were used to devise vaccination strategies and served as a basis for policymaking in many countries.” [3, p. 3]

The pro-vaccination consensus

Vaccination campaigns have had spectacular successes (smallpox, polio, tetanus, COVID-19), also through **herd immunity**.

This success underwrote a strong ethical claims during COVID-19:

- Vaccine refusal is **free-riding** [1, 2];
- Mass vaccination would **surely** improve public health.

“Transmission models were used to devise vaccination strategies and served as a basis for policymaking in many countries.” [3, p. 3]

We examine the evidential basis for “mass vaccination surely helps.”

The Claim

Our claim

There exist viral regimes in which vaccinating **only the vulnerable** produces **fewer total deaths** than vaccinating **everyone**, even when the vaccine is:

- safe (no adverse reactions, no nocebo),
- freely available in unlimited quantity,
- distributed and administered immediately,
- genuinely effective for the non-vulnerable.

Our claim

There exist viral regimes in which vaccinating **only the vulnerable** produces **fewer total deaths** than vaccinating **everyone**, even when the vaccine is:

- safe (no adverse reactions, no nocebo),
- freely available in unlimited quantity,
- distributed and administered immediately,
- genuinely effective for the non-vulnerable.

Two disclaimers, up front

This is **not** a claim about actual COVID-19, and **not** a policy recommendation. It is a **constructive counterexample** in service of **epistemic humility**.

The Model

An agent-based SISD model

Agents move by random walk on a 25×25 grid. When an infected and a susceptible agent meet, the virus is transmitted with some probability.

An agent-based SISD model

Agents move by random walk on a 25×25 grid. When an infected and a susceptible agent meet, the virus is transmitted with some probability.

The acronym **SISD** names the states an agent moves through:

- **Susceptible**: healthy, can be infected on contact.
- **Infected**: each step it either dies, recovers, or stays infected.
- **Susceptible again**: a recovered agent returns to S, now carrying added **immunity**.
- **Dead**: an absorbing state.

An agent-based SISD model

Agents move by random walk on a 25×25 grid. When an infected and a susceptible agent meet, the virus is transmitted with some probability.

The acronym **SISD** names the states an agent moves through:

- **Susceptible**: healthy, can be infected on contact.
- **Infected**: each step it either dies, recovers, or stays infected.
- **Susceptible again**: a recovered agent returns to S, now carrying added **immunity**.
- **Dead**: an absorbing state.

In short: Susceptible $\rightarrow$ Infected $\rightarrow$ {recover (back to S, with immunity) **or** die}.

Four added features

- **Vulnerable subgroup** (10%): higher infection & death probability, slower recovery [4].

Four added features

- **Vulnerable subgroup** (10%): higher infection & death probability, slower recovery [4].
- **Immunity**: each recovery lowers future infection/death probability and speeds recovery [5].

Four added features

- **Vulnerable subgroup** (10%): higher infection & death probability, slower recovery [4].
- **Immunity**: each recovery lowers future infection/death probability and speeds recovery [5].
- **Vaccination**: large reduction in infection probability, death probability, recovery time [6].

Four added features

- **Vulnerable subgroup** (10%): higher infection & death probability, slower recovery [4].
- **Immunity**: each recovery lowers future infection/death probability and speeds recovery [5].
- **Vaccination**: large reduction in infection probability, death probability, recovery time [6].
- **Viral age**, the distinctive ingredient: lethality **decreases** with each transmission (transmission-virulence trade-off).

Four added features

- **Vulnerable subgroup** (10%): higher infection & death probability, slower recovery [4].
- **Immunity**: each recovery lowers future infection/death probability and speeds recovery [5].
- **Vaccination**: large reduction in infection probability, death probability, recovery time [6].
- **Viral age**, the distinctive ingredient: lethality **decreases** with each transmission (transmission-virulence trade-off).

“... a pathogen ... will be outcompeted by mutants that cause less damage to the host and allow it to ... infect other hosts.” [7, p. 71]

Two strategies, and why S2 might win

The strategies

Strategy 1: vaccinate **all** agents. **Strategy 2:** vaccinate **only the vulnerable**.

Two strategies, and why S2 might win

The strategies

Strategy 1: vaccinate **all** agents. **Strategy 2:** vaccinate **only the vulnerable**.

Qualitative working hypothesis:

- Under S2 the virus circulates mostly among **non-vulnerable, unvaccinated** agents, who are unlikely to die.

Two strategies, and why S2 might win

The strategies

Strategy 1: vaccinate **all** agents. **Strategy 2:** vaccinate **only the vulnerable**.

Qualitative working hypothesis:

- Under S2 the virus circulates mostly among **non-vulnerable, unvaccinated** agents, who are unlikely to die.
- There it transmits often, so it **ages and turns benign faster**.

Two strategies, and why S2 might win

The strategies

Strategy 1: vaccinate **all** agents. **Strategy 2:** vaccinate **only the vulnerable**.

Qualitative working hypothesis:

- Under S2 the virus circulates mostly among **non-vulnerable, unvaccinated** agents, who are unlikely to die.
- There it transmits often, so it **ages and turns benign faster**.
- A benign virus in an unvaccinated agent can be **less deadly** than a young virus in a vaccinated one.

Two strategies, and why S2 might win

The strategies

Strategy 1: vaccinate **all** agents. **Strategy 2:** vaccinate **only the vulnerable**.

Qualitative working hypothesis:

- Under S2 the virus circulates mostly among **non-vulnerable, unvaccinated** agents, who are unlikely to die.
- There it transmits often, so it **ages and turns benign faster**.
- A benign virus in an unvaccinated agent can be **less deadly** than a young virus in a vaccinated one.

So it is possible S2 yields fewer deaths, but only simulation can tell.

Results

Searching the parameter space

We fix most parameters and vary three: **viral ageing**, **vaccination effect**, **death probability**.

- Bayesian optimisation (`scikit-optimize`, GP regression) [8].
- Objective: average death difference $S1 - S2$, over $n = 200$ random seeds per setting.
- $k = 1000$ settings explored $\Rightarrow$ **224** where $S2$ beats $S1$.

Searching the parameter space

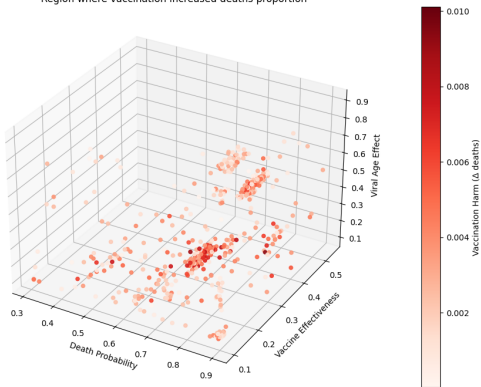
We fix most parameters and vary three: **viral ageing**, **vaccination effect**, **death probability**.

- Bayesian optimisation (`scikit-optimize`, GP regression) [8].
- Objective: average death difference $S1 - S2$, over $n = 200$ random seeds per setting.
- $k = 1000$ settings explored $\Rightarrow$ **224** where $S2$ beats $S1$.

Goal: not that such regimes are common, but that they exist, and are spread out.

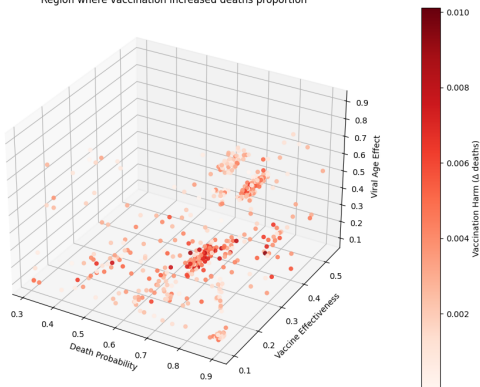
S2 causes fewer deaths than S1

Region where vaccination increased deaths proportion



S2 causes fewer deaths than S1

Region where vaccination increased deaths proportion



Two lessons:

- the advantage is **small** (max ~1% of agents on average), and
- the regions are **scattered**: no tidy rule for when S1 wins.

The ethical inversion

To minimise **total** deaths, it is sometimes best for the majority to **raise** their own risk of infection and death.

The ethical inversion

To minimise **total** deaths, it is sometimes best for the majority to **raise** their own risk of infection and death.

Free-riding → public service

Foregoing vaccination lets the virus attenuate in a low-risk population, a **negative herd immunity** effect [9] that benefits everyone. The “free-rider” is in fact taking on risk for the group.

The ethical inversion

To minimise **total** deaths, it is sometimes best for the majority to **raise** their own risk of infection and death.

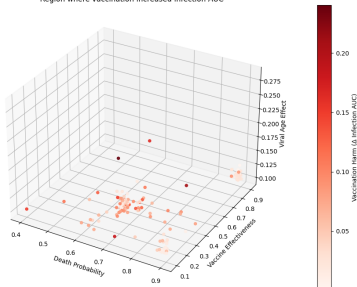
Free-riding → public service

Foregoing vaccination lets the virus attenuate in a low-risk population, a **negative herd immunity** effect [9] that benefits everyone. The “free-rider” is in fact taking on risk for the group.

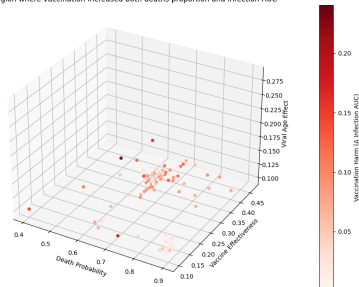
At times, the needs of the few **do** outweigh the needs of the many, if measured by simple averaging.

Beyond deaths: area under the curve

Region where vaccination increased infection AUC

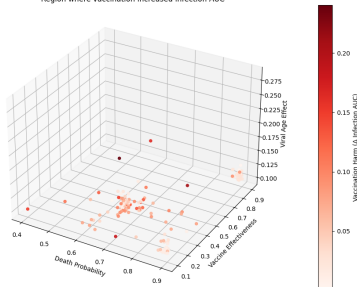


Region where vaccination increased both deaths proportion and infection AUC

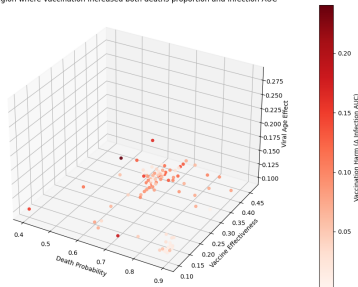


Beyond deaths: area under the curve

Region where vaccination increased infection AUC

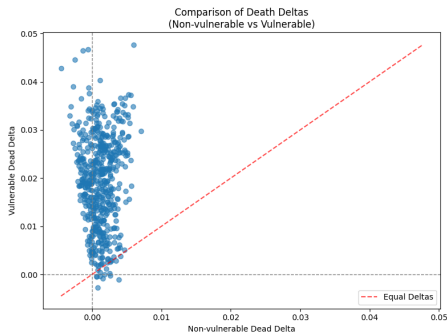


Region where vaccination increased both deaths proportion and infection AUC



Left: settings where S2 also has the smaller infection burden (an economic proxy). **Right:** settings where S2 wins on **both** deaths and AUC. Fewer deaths $\neq$ smaller AUC.

Who gains from Strategy 2?

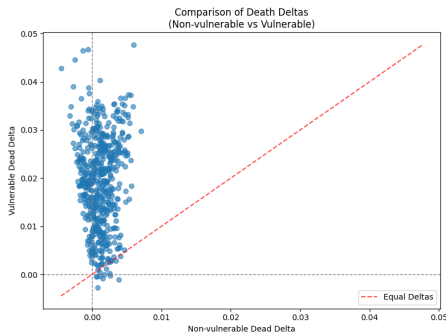


Each dot: one of the 224 settings.

x: S2 advantage for **non-vulnerable**.

y: S2 advantage for **vulnerable**.

Who gains from Strategy 2?



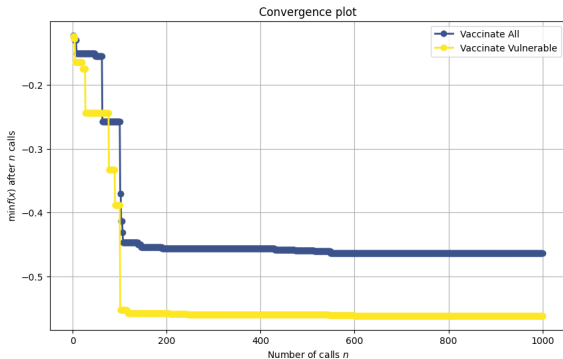
Each dot: one of the 224 settings.

x : S2 advantage for **non-vulnerable**.

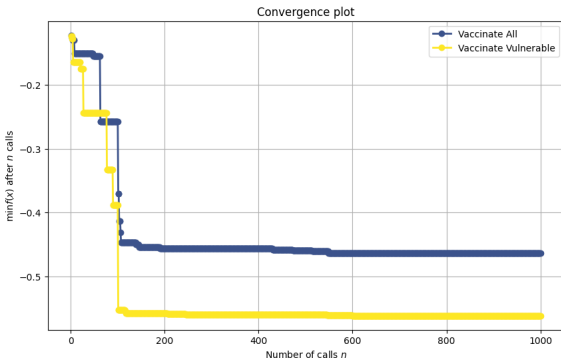
y : S2 advantage for **vulnerable**.

Top-left ($x < 0$, $y > 0$): the non-vulnerable majority **sacrifices**, the vulnerable minority gains more: net benefit overall.

Worst cases cut the other way

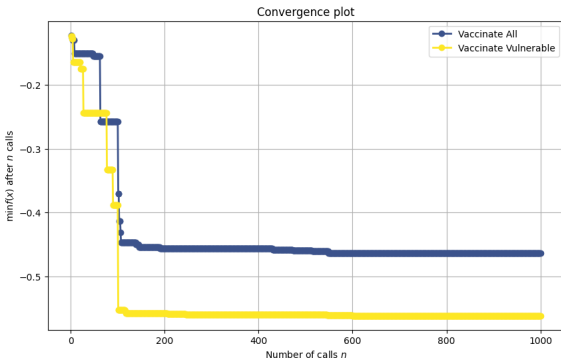


Worst cases cut the other way



Searching **independently** for the deadliest setting of each strategy: universal vaccination's worst case (47% dead) beats selective's (56%).

Worst cases cut the other way



Searching **independently** for the deadliest setting of each strategy: universal vaccination's worst case (47% dead) beats selective's (56%).

A strongly harm-averse planner under deep uncertainty may rationally prefer S1.

Objection & Humility

“Your model is just speculative”

Objection: established COVID models all favoured universal vaccination: why trust this simpler one?

“Your model is just speculative”

Objection: established COVID models all favoured universal vaccination: why trust this simpler one?

- We do **not** model COVID-19; we aim at **future** pandemics.

“Your model is just speculative”

Objection: established COVID models all favoured universal vaccination: why trust this simpler one?

- We do **not** model COVID-19; we aim at **future** pandemics.
- Simplicity enables understanding and **robustness checks** large black-box models cannot run.

“Your model is just speculative”

Objection: established COVID models all favoured universal vaccination: why trust this simpler one?

- We do **not** model COVID-19; we aim at **future** pandemics.
- Simplicity enables understanding and **robustness checks** large black-box models cannot run.
- Model **agreement** may reflect a **shared missing mechanism** (viral ageing), not truth.

“Your model is just speculative”

Objection: established COVID models all favoured universal vaccination: why trust this simpler one?

- We do **not** model COVID-19; we aim at **future** pandemics.
- Simplicity enables understanding and **robustness checks** large black-box models cannot run.
- Model **agreement** may reflect a **shared missing mechanism** (viral ageing), not truth.
- One anomalous model raises the prospect of **many**.

“Your model is just speculative”

Objection: established COVID models all favoured universal vaccination: why trust this simpler one?

- We do **not** model COVID-19; we aim at **future** pandemics.
- Simplicity enables understanding and **robustness checks** large black-box models cannot run.
- Model **agreement** may reflect a **shared missing mechanism** (viral ageing), not truth.
- One anomalous model raises the prospect of **many**.

Historical precedent: Marek's disease [10] and pneumococcal serotype replacement [11], vaccination with real negative evolutionary outcomes.

Limits of ethical inference

What the model does **not** settle:

- It uses **mortality only**, ignoring morbidity, long COVID, liberty, economic cost, fairness.

Limits of ethical inference

What the model does **not** settle:

- It uses **mortality only**, ignoring morbidity, long COVID, liberty, economic cost, fairness.
- It targets **population strategies**, not the morality of **individual** acts.

Limits of ethical inference

What the model does **not** settle:

- It uses **mortality only**, ignoring morbidity, long COVID, liberty, economic cost, fairness.
- It targets **population strategies**, not the morality of **individual** acts.
- Its ethics ride on **contestable** assumptions about viral evolution and immunity.

Limits of ethical inference

What the model does **not** settle:

- It uses **mortality only**, ignoring morbidity, long COVID, liberty, economic cost, fairness.
- It targets **population strategies**, not the morality of **individual** acts.
- Its ethics ride on **contestable** assumptions about viral evolution and immunity.

Underdetermination

Even a coherent, empirically motivated model can leave the **ethical** conclusion substantially underdetermined.

Conclusions

- Our grasp of vaccination outcomes is, in many situations, **too poor** to support strong moral verdicts either way.

Conclusions

- Our grasp of vaccination outcomes is, in many situations, **too poor** to support strong moral verdicts either way.
- We should be cautious before labelling vaccine hesitancy as free-riding, a boon, or negligence.

Conclusions

- Our grasp of vaccination outcomes is, in many situations, **too poor** to support strong moral verdicts either way.
- We should be cautious before labelling vaccine hesitancy as free-riding, a boon, or negligence.
- Ethical **pluralism** only sharpens the point: different theories, different verdicts.

Conclusions

- Our grasp of vaccination outcomes is, in many situations, **too poor** to support strong moral verdicts either way.
- We should be cautious before labelling vaccine hesitancy as free-riding, a boon, or negligence.
- Ethical **pluralism** only sharpens the point: different theories, different verdicts.

The remedy is epistemic humility and honest uncertainty communication, itself a contribution to pandemic preparedness.

Conclusion

Takeaways

Takeaways

- 1 ***A constructive counterexample:*** under favourable assumptions, selective vaccination can still minimise deaths.

Takeaways

- 1 **A constructive counterexample:** under favourable assumptions, selective vaccination can still minimise deaths.
- 2 **A neglected mechanism:** the transmission-virulence trade-off (viral ageing) changes the ethical picture.

Takeaways

- 1 **A constructive counterexample:** under favourable assumptions, selective vaccination can still minimise deaths.
- 2 **A neglected mechanism:** the transmission-virulence trade-off (viral ageing) changes the ethical picture.
- 3 **A stance:** strong moral conclusions drawn straight from epidemiological models deserve humility under deep uncertainty.

Thanks!

Many thanks for listening!

Thanks!

Many thanks for listening!

Questions?

juergen_landes@yahoo.de

Appendix: Model parameters

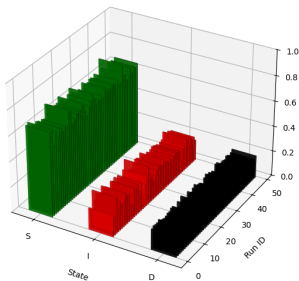
- $N = 500$ agents, 25×25 grid, 10% initially infected, 10% vulnerable.
- Base infection prob. 0.25; base death prob. 0.01; recovery time 30.
- Vulnerability penalty +300%; immunity effect 10%/recovery.
- Vaccination effect 70%; viral ageing 10%/transmission.
- Optimisation: $2 \times 200 \times 1000 = 400,000$ simulations, 11.5h on TPU.

Appendix: Death probability update

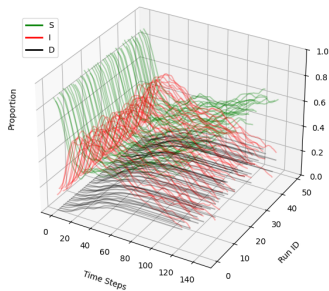
```
1 if self.vaxxed:
2     base_death_prob = self.death_prob * (1 - self.vax_effect)
3 else:
4     base_death_prob = self.death_prob
5
6 base_death_prob = max(
7     base_death_prob *
8     (1 - self.viral_age_effect * self.viral_age
9     - self.immune_adaptation_effect * self.immunity_level),
10    0)
11
12 if self.rng.random() < base_death_prob:
13     self.state = 'D'
```

Appendix: An example single run (S2)

Final Agent State Proportions (3D Bar)
Vaccinate Vulnerable



Smoothed State Proportions Over Time (3D)
Vaccinate Vulnerable



Susceptible (green), infected (red), dead (black). Virus becomes endemic but harmless via viral ageing and immunity growth.

References I

- Mariëtte van den Hoven. "Why One Should Do One's Bit: Thinking about Free Riding in the Context of Public Health Ethics". In: **Public Health Ethics** 5.2 (2012), pp. 154–160. URL: <https://doi.org/10.1093/phe/phs023>.
- Alberto Giubilini, Thomas Douglas, and Julian Savulescu. "The moral obligation to be vaccinated: utilitarianism, contractualism, and collective easy rescue". In: **Medicine, Health Care and Philosophy** 21.4 (2018), pp. 547–560. URL: <https://doi.org/10.1007/s11019-018-9829-y>.
- Igor M. Rouzine and Ganna Rozhnova. "Evolutionary implications of SARS-CoV-2 vaccination for the future design of vaccination strategies". In: **Communications Medicine** 3.1 (2023). URL: <https://doi.org/10.1038/s43856-023-00320-x>.
- Mohitosh Biswas et al. "Association of Sex, Age, and Comorbidities with Mortality in COVID-19 Patients: A Systematic Review and Meta-Analysis". In: **Intervirology** 64.1 (2021), pp. 36–47. URL: <https://doi.org/10.1159/000512592>.
- VJ Hall, J Victoria, S. Foulkes et al. and the SIREN Study Group. "SARS-CoV-2 infection rates of antibody-positive compared with antibody-negative health-care workers in England: a large, multicentre, prospective cohort study (SIREN)". In: **The Lancet** 397.10283 (2021), pp. 1459–1469. URL: [https://doi.org/10.1016/s0140-6736\(21\)00675-9](https://doi.org/10.1016/s0140-6736(21)00675-9).
- Daniel M. Altmann and Rosemary J. Boyton. "COVID-19 vaccination: The road ahead". In: **Science** 375.6585 (2022), pp. 1127–1132. URL: <https://doi.org/10.1126/science.abn1755>.
- Ádám Kun et al. "Do pathogens always evolve to be less virulent? The virulence–transmission trade-off in light of the COVID-19 pandemic". In: **Biologia Futura** 74.1–2 (2023), pp. 69–80. URL: <https://doi.org/10.1007/s42977-023-00159-2>.
- Tim Head et al. **scikit-optimize/scikit-optimize**. Version v0.9.0. 2021. DOI: 10.5281/zenodo.5565057. URL: <https://doi.org/10.5281/zenodo.5565057>.
- J. Luyten et al. "Vaccination Policy and Ethical Challenges Posed by Herd Immunity, Suboptimal Uptake and Subgroup Targeting". In: **Public Health Ethics** 4.3 (2011), pp. 280–291. URL: <https://doi.org/10.1093/phe/phr032>.
- Andrew F. Read et al. **Data from: Imperfect vaccination can enhance the transmission of highly virulent pathogens**. 2015. URL: <https://doi.org/10.5061/dryad.4tn48>.

References II

Germaine Hanquet et al. "Serotype Replacement after Introduction of 10-Valent and 13-Valent Pneumococcal Conjugate Vaccines in 10 Countries, Europe". In: **Emerging Infectious Diseases** 28.1 (2022), pp. 137–138. URL: <https://doi.org/10.3201/eid2801.210734>.